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Congenetic hematopoietic stem cell transplantation promotes survival of heart allografts in murine models of acute and chronic rejection

Hassan Sadozai, Vanessa Rojas-Luengas, Kaveh Farrokhi, Sajad Moshkelgosha, Qinli Guo, Wei He, Angela Li, Jianhua Zhang, Conan Chua, Dario Ferri, Muhtashim Mian, Oyedele Adeyi, Michael Seidman, Reginald M Gorczynski, Stephen Juvet, Harold Atkins, Gary A Levy, Andrzej Chruscinski

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Abstract

The ability to induce tolerance would be a major advance in the field of solid organ transplantation. Here, we investigated whether autologous (congenic) hematopoietic stem cell transplantation (HSCT) could promote tolerance to heart allografts in mice. In an acute rejection model, fully MHC-mismatched BALB/c hearts were heterotopically transplanted into C57BL/6 (CD45.2) mice. One week later, recipient mice were lethally irradiated and reconstituted with congenic B6 CD45.1 Lin[−]Sca1⁺ckit⁺ cells. Recipient mice received a 14-day course of rapamycin both to prevent rejection and to expand regulatory T cells (Tregs). Heart allografts in both untreated and rapamycin-treated recipients that did not undergo HSCT were rejected within 33 days (median survival time = 8 days for untreated recipients, median survival time = 32 days for rapamycin-treated recipients), whereas allografts in HSCT-treated recipients had a median survival time of 55 days (P < 0.001 vs. both untreated and rapamycin-treated recipients). Enhanced allograft survival following HSCT was associated with increased intragraft Foxp3⁺ Tregs, reduced intragraft B cells, and reduced serum donor-specific antibodies. In a chronic rejection model, Bm12 hearts were transplanted into C57BL/6 (CD45.2) mice, and congenic HSCT was performed two weeks following heart transplantation. HSCT led to enhanced survival of allografts (median survival time = 70 days vs. median survival time = 28 days in untreated recipients, P < 0.01). Increased allograft survival post-HSCT was associated with prevention of autoantibody development and absence of vasculopathy. These data support the concept that autologous HSCT can promote immune tolerance in the setting of allotransplantation. Further studies to optimize HSCT protocols should be performed before this procedure is adopted clinically.

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- transplantation

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KW - rejection

KW - stem cells

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